DW binary systems

Sources:

PanStar-1 image access, https://ps1images.stsci.edu/cgi-bin/ps1cutouts?pos=18%3A32%3A36.539+++20%3A31%3A08.202&filter=color&filter=g&filter=r&filter=i&filter=z&filter=y&filetypes=stack&auxiliary=data&size=240&output_size=0&verbose=0&autoscale=99.500000&catlist= :

- Got the images and most of the description from the far right image (in the infrared) that was loaded when inputting the RA and Dec of the stars Manuel gave me

https://aladin.cds.unistra.fr/AladinLite/

- Got some of the extra data such as the Gaia and Simba info and also got the photo for J0338-8139

https://simbad.cds.unistra.fr/simbad/sim-basic?Ident=23%3A22%3A30.203++++5%3A09%3A42.058&submit=SIMBAD+search

5-minute talk, background and reason for this research:

- Script

Download all the tables and create my own plots excluding the spikes

Seemly good candidates 🤔:

Mostly Isolated

1), 2), 10)(

1) J2322+0509, (2) J0935+4411(very bright), (10)J1235+1543(kind of looks like it's in the void)

Has one or two (faint)

9), 11)

(9) J1506-1125, (11)J0056-0611(some friend with it but seems good distance)

Has one or two (bright)

4), 6), 7), 13), 14)(

4) J0634+3803(maybe), (7) J1526-2711(also maybe), (13)J1436+5010(some bright star right on the edge)

Lot of other (faint with maybe 1 or 2 bright)

3), 12)

(12) J0923+3028(has some buddies also only one not found in PanSTAR)

Bright really really close by

3)

Lot of other (bright)

5), 8)

Name Gmag Gmag_e ra dec

1) J2322+0509 *

18.7466 0.0020 23:22:30.203 5:09:42.058

- Doesn't have too many other sources, maybe some really dim star near by



- Info from Aladin
  - Gaia DR3 26607734, PM:13.667 (from aladdin website where i can click on the little circle and it tells me the Gaia data if Gaia is selected)
  - Gaia RA and DEC: 350.62584588574 5.16168267182
  - Main ID from Simbad catalog: PB 5353, PM: pmra11.758 pmdec-6.965
  - Sinbad RA and Dec: 350.6257934130 +5.1617136283
- Simbad Coord website
  - ID: PB 5353
  - PM: **11.758 -6.965**
- Using python code
  - Calculated RA (in degrees): 350.625898
  - Calculated Dec (in degrees): +5.161652
- Compared to Gaia

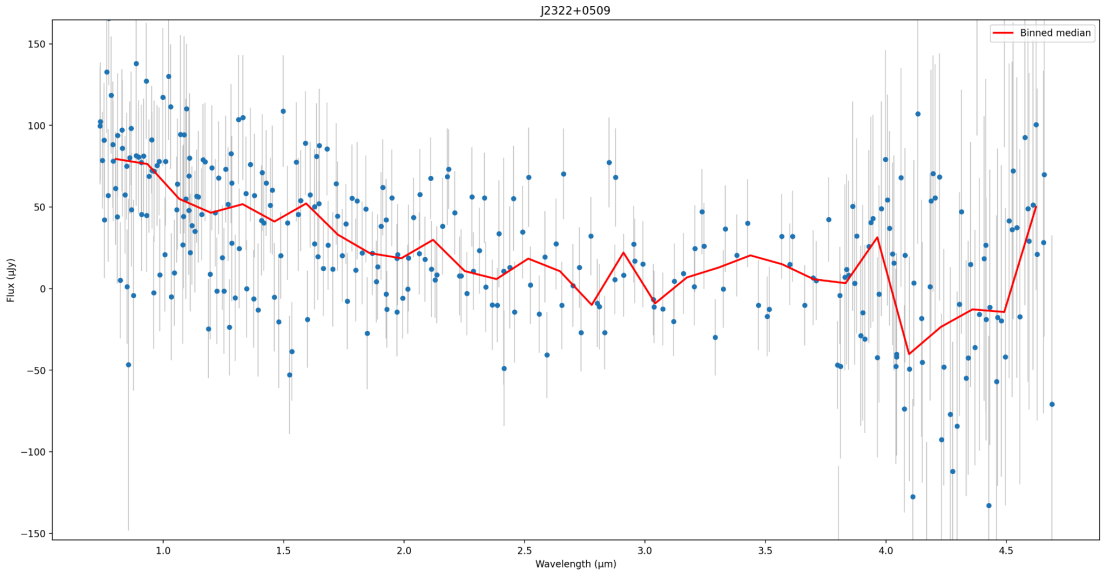
Full	r arcmin	RAJ2000	DECJ2000	USNO-A1.0	RA_ICRS	DEC_ICRS
1	0.2810	23 22 00.72218	+05 09 12.9526	0900-20539545	350.503009	+05.153598

1	0.0883	23 22 00.3748817259	+05 09 05.786658354	0.0876	350.50113214234	+05.15094486519

Photometry plot found on ISRA to be at 0?

1) J2322+0509 *

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



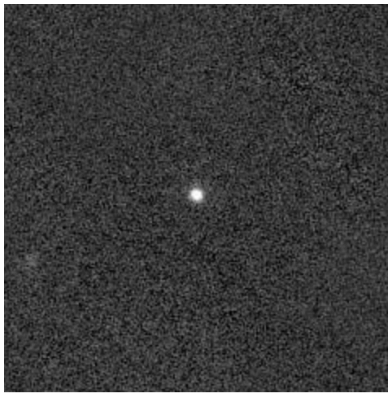
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

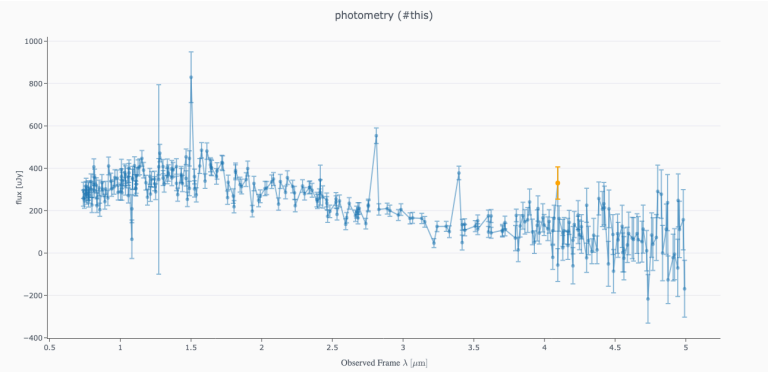
2) J0935+4411 *

17.8045 0.0062 9:35:06.897 44:11:06.919

- Doesn't have really any stars near by



- Info from Aladin
  - Gaia DR3 81509298, PM:20.351
  - Gaia RA and DEC: 143.77873662778 44.18525527397
  - Main ID from Simbad catalog: US 738, PM: pmra-20.315 pmdec-1.213
  - Sinbad RA and Dec: 143.7788625382 +44.1852606652
- Simbad Coord website
  - ID: US 738
  - PM: **-20.315 -1.213**
- Using python and the plotting with IRSA
  - Calculated RA (in degrees): 143.625898
  - Calculated Dec (in degrees): 44.185250

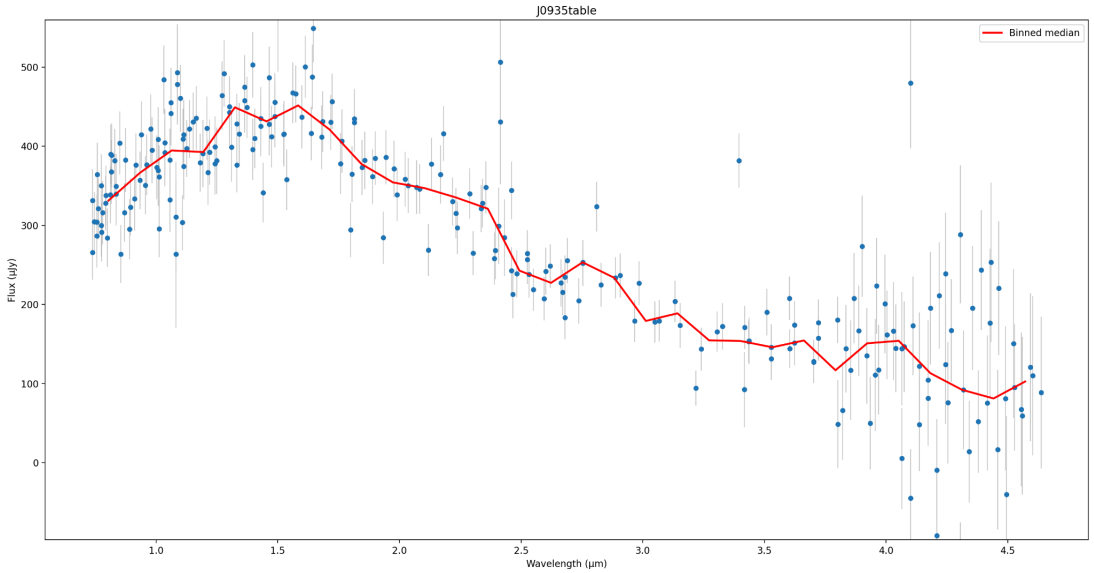


- Compared to Gaia

Full	Dr3 Name	PM	RA_PM	DEC_PM	RA_ICRS	DEC_ICRS
1	Gaia DR3 815092369380832512	15.626	-14.952 ~ -3.231	-4.542	143.76883289705	+44.15826616238

2) J0935+4411 *

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



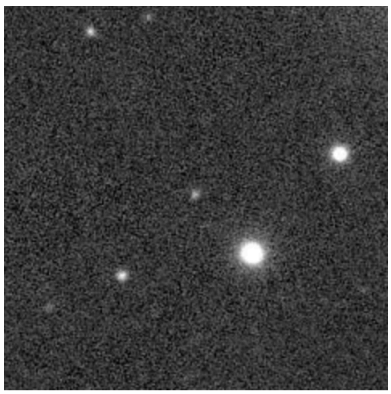
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down. And a nicer curve than that of ISRA but still similar

Name Gmag Gmag_e ra dec

3) J1239-2041 *

18.9418 0.0019 12:39:50.370 -20:41:42.276

- In between two other stars and a third star above it (so 4 total sources in a 60 parsecs pixel)

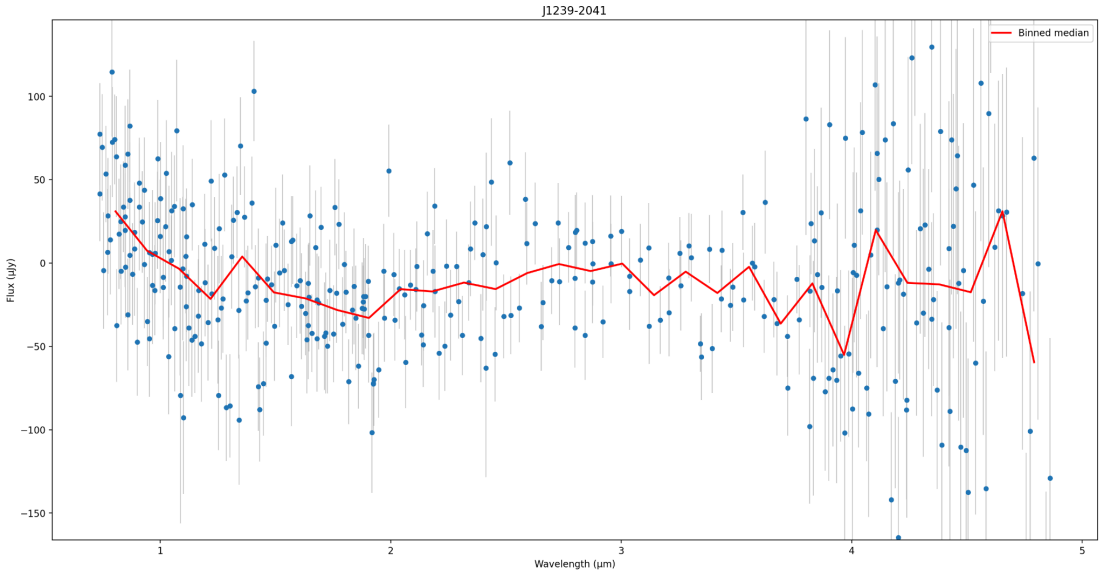


- Info from Aladin
  - Gaia DR3 350361328, PM:11.904
  - Gaia RA and DEC: 189.95987569252 -20.69507653573
  - Main ID from Simbad catalog: GALEX J123950.2-2 PM: pmra-9.039 pmdec7.746
  - Sinbad RA and Dec: 189.9599186377 -20.6951109628
- Simbad Coord website
  - ID: GALEX J123950.2-2
  - PM: **-9.039 7.746**
- Using python code
  - Calculated RA (in degrees): 189.959832
  - Calculated Dec (in degrees): -20.695042
- Compared to Gaia
- Found 2026-09-18 22:40:01
- 2026-09-18 22:40:06

Photometry plot found on ISRA to be at 0?

J1239-2041 *

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



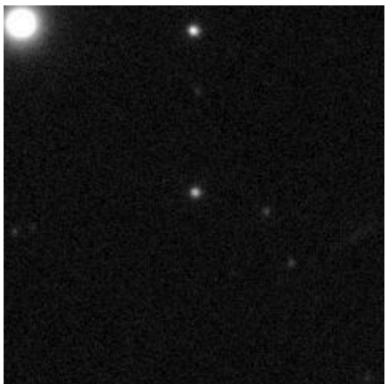
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

4) J0634+3803

17.1331 0.0011 6:34:49.904 38:03:51.975

- One big ball of light in the corner of the square and the many target looks to be significantly dimmer than the big ball of light

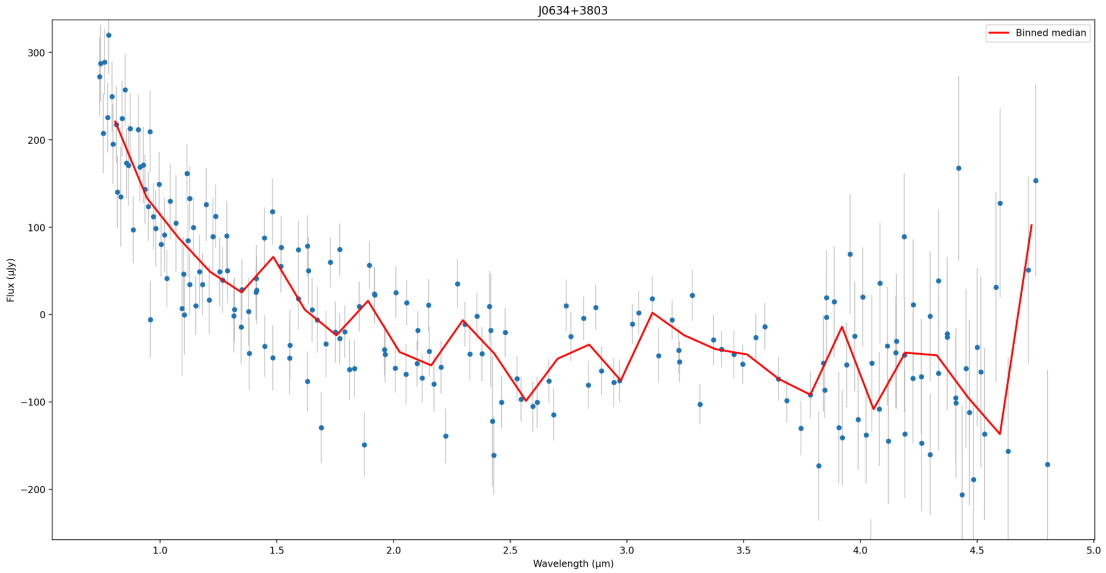


- Info from Aladin
  - Gaia DR3 943429565, PM:37.951
  - Gaia RA and DEC: 98.70793535266 38.06443759899
  - Main ID from Simbad catalog: SDSS J063449.92+3, PM: pmra-23.590 pmdec-29.729
  - Sinbad RA and Dec: 98.7080685180 +38.0645697262
- Simbad Coord website
  - ID: SDSS J063449.92+3
  - PM: **-23.590 -29.729**
- Using python code
  - Calculated RA (in degrees): 98.707800
  - Calculated Dec (in degrees): 38.064305
- Compared to Gaia

Photometry plot found on ISRA to be at 0?

J0634+3803

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



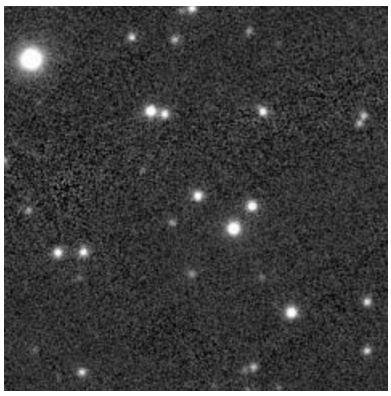
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

5) J1946-1318

17.4439 0.0015 19:46:12.798 -13:18:22.786

- A lot of different stars

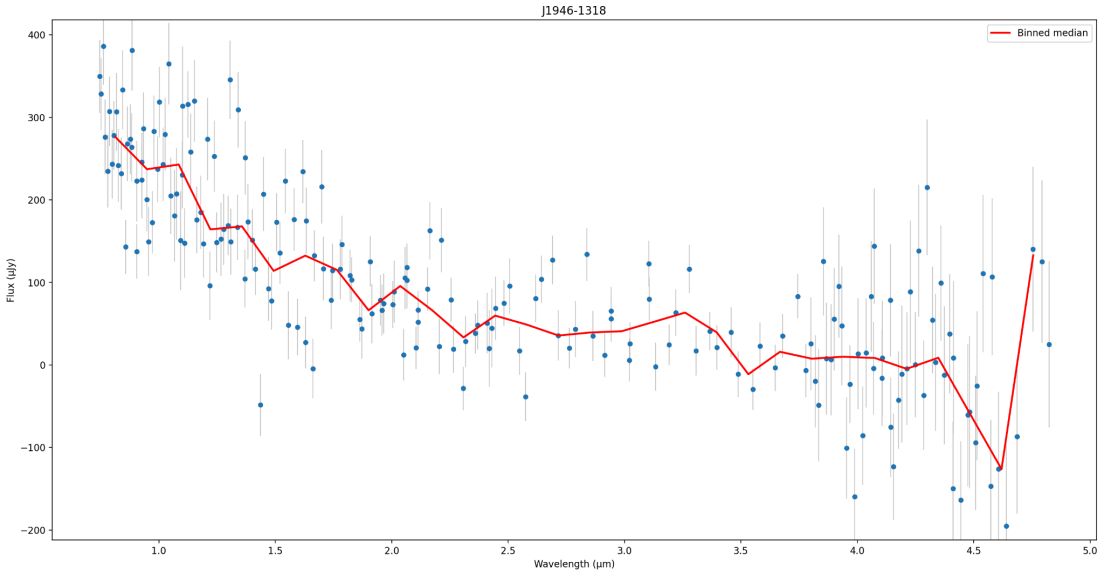


- Info from Aladin
  - Gaia DR3 418346750, PM:4.679
  - Gaia RA and Dec: 296.55332409174 -13.30632954976
  - Main ID from Simbad catalog: GALEX J194612.7-1, PM: pmra4.619 pmdec0.744
  - Sinbad RA and Dec: 296.5533029970 -13.3063328585
- Simbad Coord website
  - ID: GALEX J194612.7-1
  - PM: **4.619 0.744**
- Using python code
  - Calculated RA (in degrees): 296.553346
  - Calculated Dec (in degrees): -13.306326
- Compared to Gaia

Photometry plot found on ISRA to be at 0?

4) J0634+3803

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



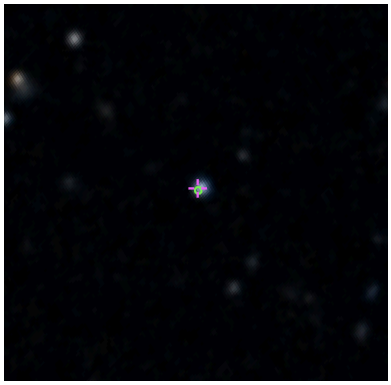
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

6) J0338-8139

17.2902 0.0011 3:38:16.193 -81:39:29.967

- The PanSTARRS 3PI survey images cover the sky north of declination -30 deg.
- No result cant see withe PanStar tool
- Due to me for getting which thing to set aladdin to i dont know if its and image in the same wavelength as the rest

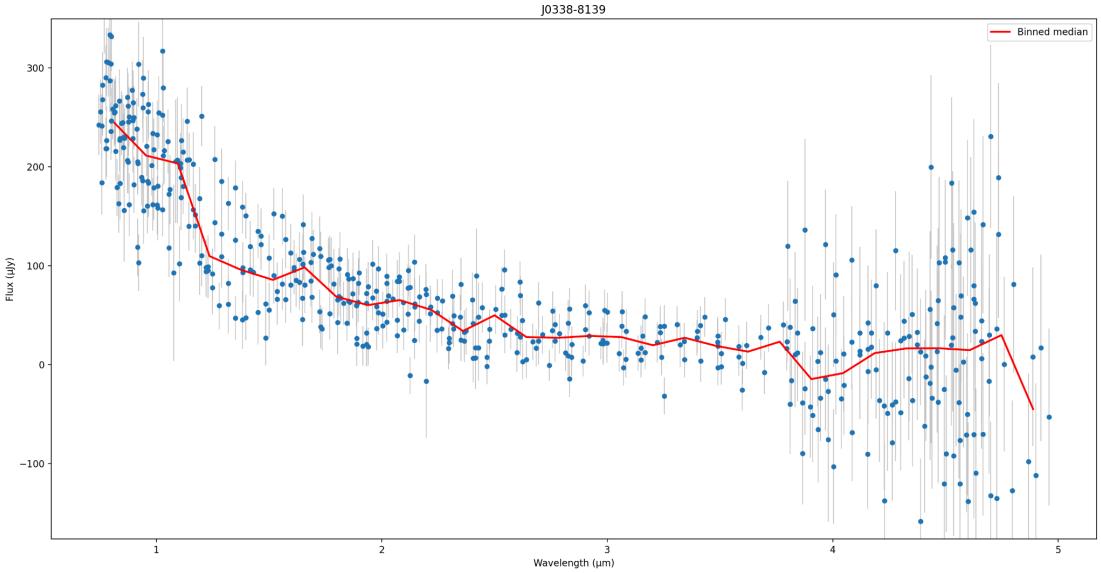


- However there is a small star right next to it
- Info from Aladin
  - Gaia DR3 46190741 PM:14.180, pmra13.029 pmdec5.596
  - Gaia RA and DEC: 54.56746987594 -81.65832415289
  - Main ID from Simbad catalog: GALEX J033815.9-8, PM: pmra13.029 pmdec5.596
  - Sinbad RA and Dec: 54.5670707387 -81.6583490236
- Simbad Coord website
  - ID: GALEX J033815.9-8
  - PM: **13.029 5.596**
- Using python code
  - Calculated RA (in degrees): 54.567870
  - Calculated Dec (in degrees): -81.658299
- Compared to Gaia

Photometry plot found on ISRA to be at 0?

6) J0338-8139

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



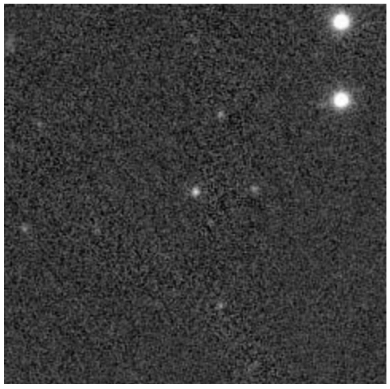
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

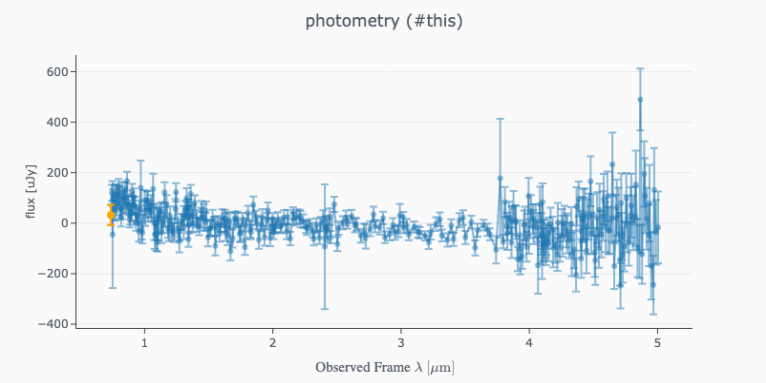
7) J1526-2711

18.3489 0.0020 15:26:01.115 -27:11:56.660

- Two little big balls of light in the corner
- Some dimmer stars close to it



- Info from Aladin
  - Gaia DR3 62136199, PM:3.482
  - Gaia RA and DEC: 231.50464756546 -27.19907220393
  - Main ID from Simbad catalog: PSO J231.5046-27, PM: pmra2.668 pmd-2.237
  - Sinbad RA and Dec: 231.5046342310 -27.1990622622
- Simbad Coord website
  - ID: PSO J231.5046-27
  - PM: **2.668 -2.237**
- Using python and the plotting with IRSA
  - Calculated RA (in degrees): 231.504659
  - Calculated Dec (in degrees): -27.199082

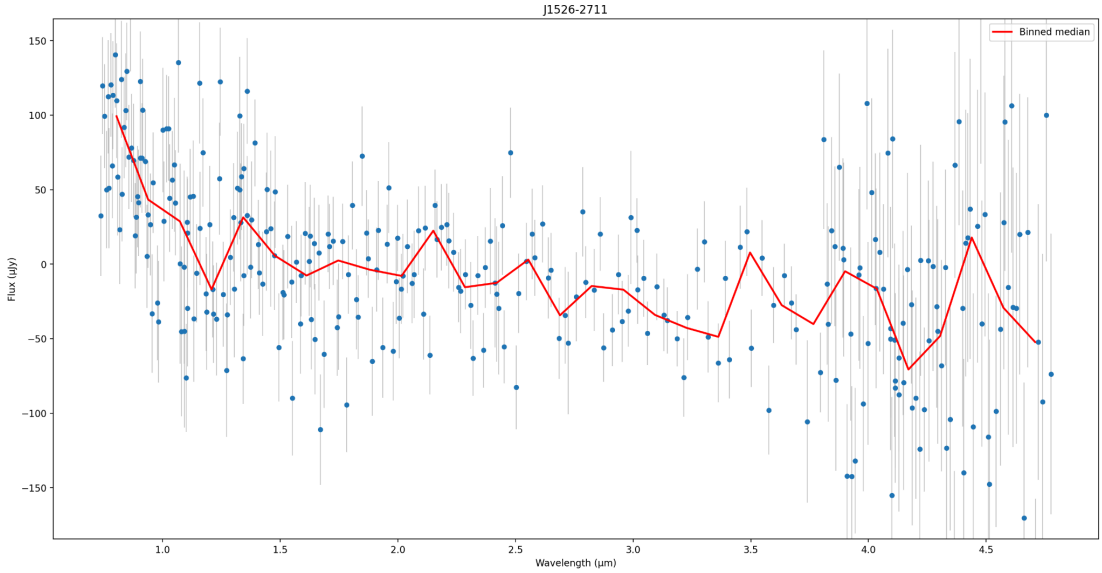


- Compared to Gaia

Weird???

7) J1526-2711

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



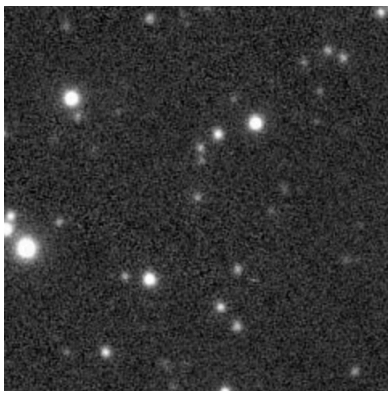
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down. Mostly in the zero like in the ISRA plot

Name Gmag Gmag_e ra dec

8) J2049+3351

18.7010 0.0024 20:49:51.274 33:51:53.126

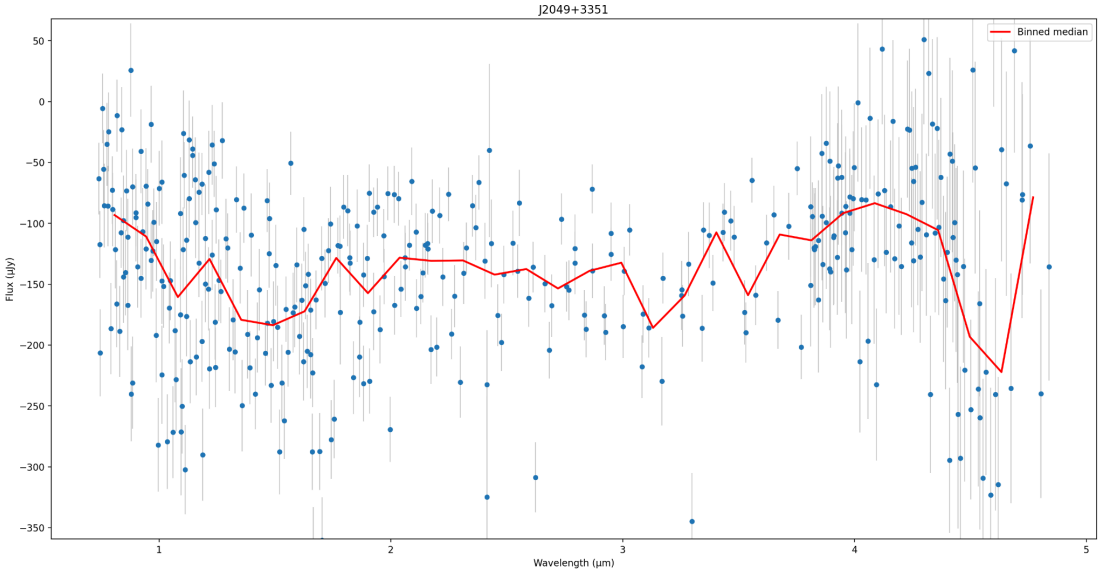
- More stars close to it



- Info from Aladin
  - Gaia DR3 18691112, PM:2.221
  - Gaia RA and DEC: 312.46364358406 33.86475720324
  - Main ID from Simbad catalog: PSO J312.4636+33 PM: pmra-0.446 pmdec-2.176
  - Sinbad RA and Dec: 312.4636459725 +33.8647668731
- Simbad Coord website
  - ID: J312.4636+33
  - PM: **-0.446 -2.176**
- Using python code
  - Calculated RA (in degrees): 312.463639
  - Calculated Dec (in degrees): 33.864748
- Compared to Gaia

8) J2049+3351

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

9) J1506-1125

17.0104 0.0015 15:06:12.345 -11:25:11.994

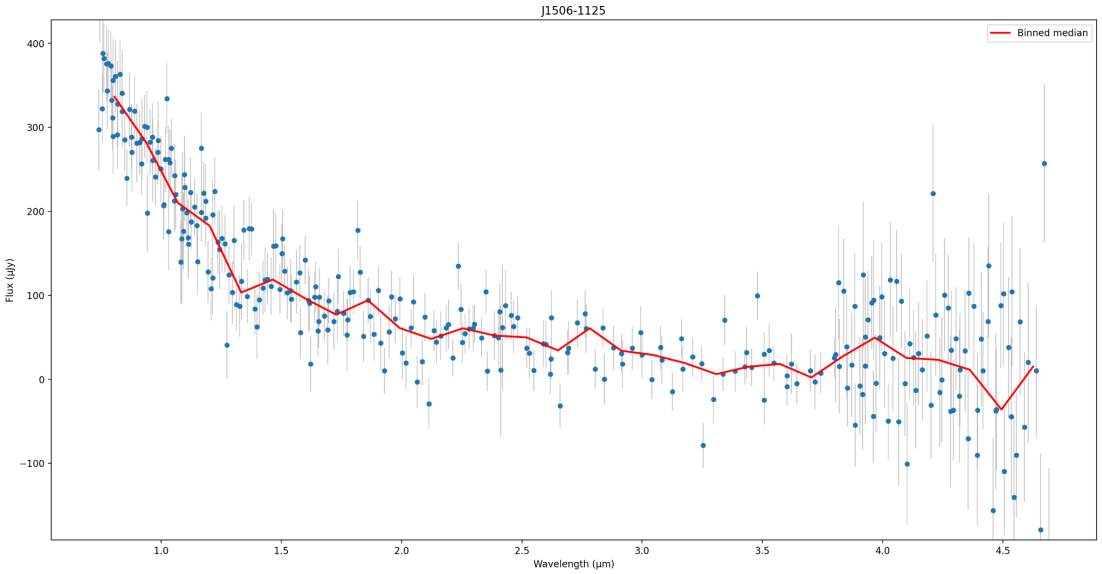
- One main star with a few little spots that seems to show in this wavelength with a small pair below and one above



- Info from Aladin
  - Gaia DR3 63128379, PM:10.720
  - Gaia RA and DEC: 226.55143810657 -11.41999826588
  - Main ID from Simbad catalog: GALEX J150612.3-1, PM: pmra0.419 pmdec-10.712
  - Sinbad RA and Dec: 226.5514362082 -11.4199506579
- Simbad Coord website
  - ID: GALEX J150612.3-1
  - PM: **0.419 -10.712**
- Using python code
  - Calculated RA (in degrees): 226.551439
  - Calculated Dec (in degrees): -11.420046
- Compared to Gaia

9) J1506-1125

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



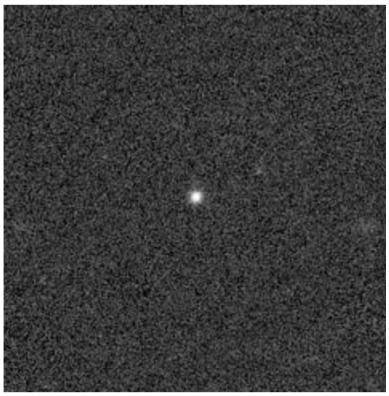
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

10) J1235+1543

17.5162 0.0012 12:35:49.853 15:43:18.954

- Single source



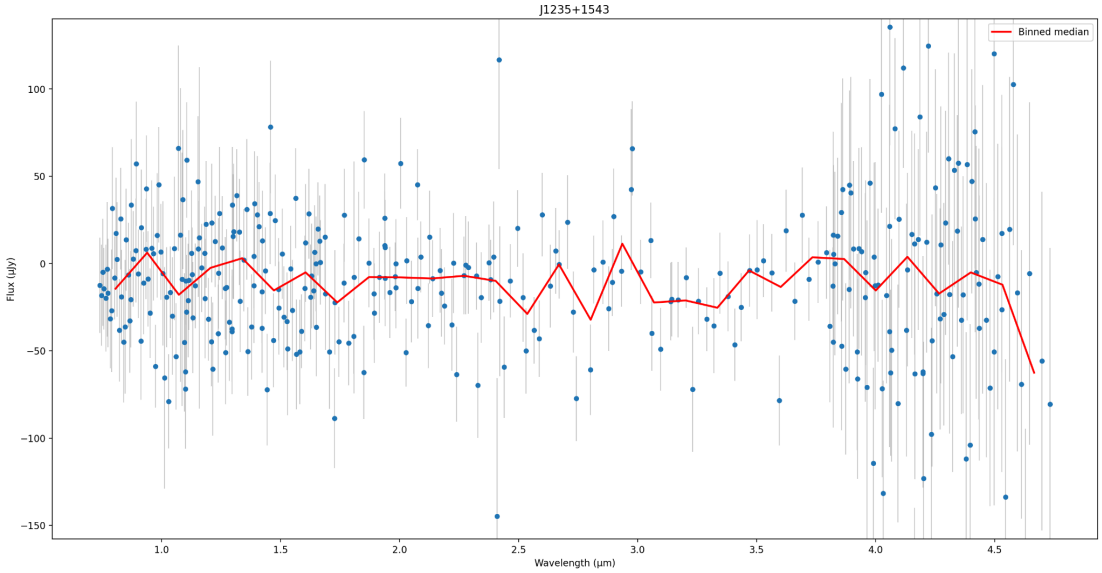
- Info from Aladin
  - Gaia DR3 39335578, PM:41.310
  - Gaia RA and DEC: 188.95771890669 15.72193179104
  - Main ID from Simbad catalog: A3 476 PM: pmra-32.511 pmdec-25.486
  - Sinbad RA and Dec: 188.9578690150 +15.7220450636
- Simbad Coord website
  - ID: A3 476
  - PM: **-32.511 -25.486**
- Using python code
  - Calculated RA (in degrees): 188.857603
  - Calculated Dec (in degrees): +15.721758
  - Nothing in the location on IRSA
- Compared to Gaia

Full	r arcmin	RAJ2000	DEJ2000	USNO-A1.0	RA_ICRS	DEC_ICRS
1	0.3618	12 35 00.68350	+15 43 19.3330	1050-06847536	188.752848	+15.722037

  - RA off by .1 but the DEC is closer, maybe test out the ISRA

10) J1235+1543

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

11) J0056-0611

17.5441 0.0010 0:56:48.240 -6:11:41.804

- Just a single other star near by

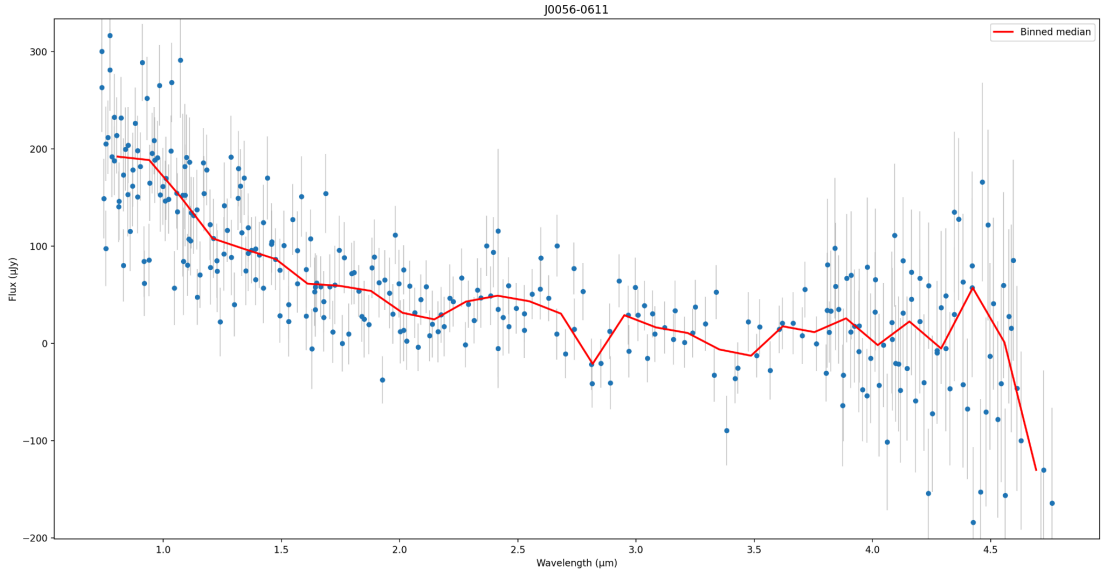


- Info from Aladin
  - Gaia DR3 25243900, PM:22.733
  - Gaia RA and DEC: 14.20099900190 -6.19494550092
  - Main ID from Simbad catalog: GALEX J005648.1-0 PM: pmra16.746 pmdec-15.374
  - Sinbad RA and Dec: 14.2009241372 -6.1948771727
- Simbad Coord website
  - ID: GALEX J005648.1-0
  - PM: **16.746 -15.374**
- Using python code
  - Calculated RA (in degrees): 14.201075
  - Calculated Dec (in degrees): -6.195014
- Compared to Gaia

Photometry plot found on ISRA to be at 0?

11) J0056-0611

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



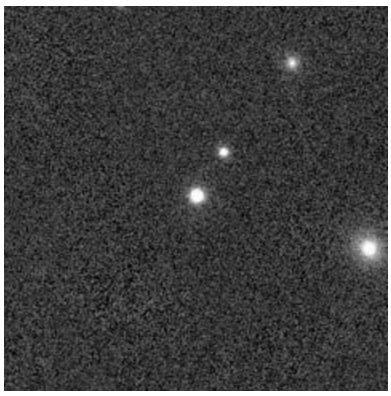
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

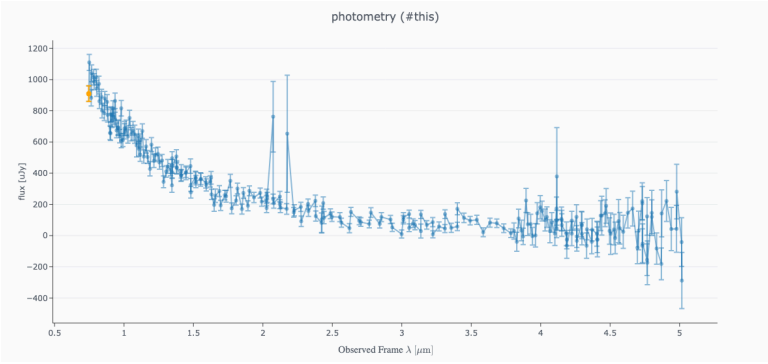
12) J0923+3028

15.9203 0.0008 9:23:45.598 30:28:04.609

- 3 other stars in the image



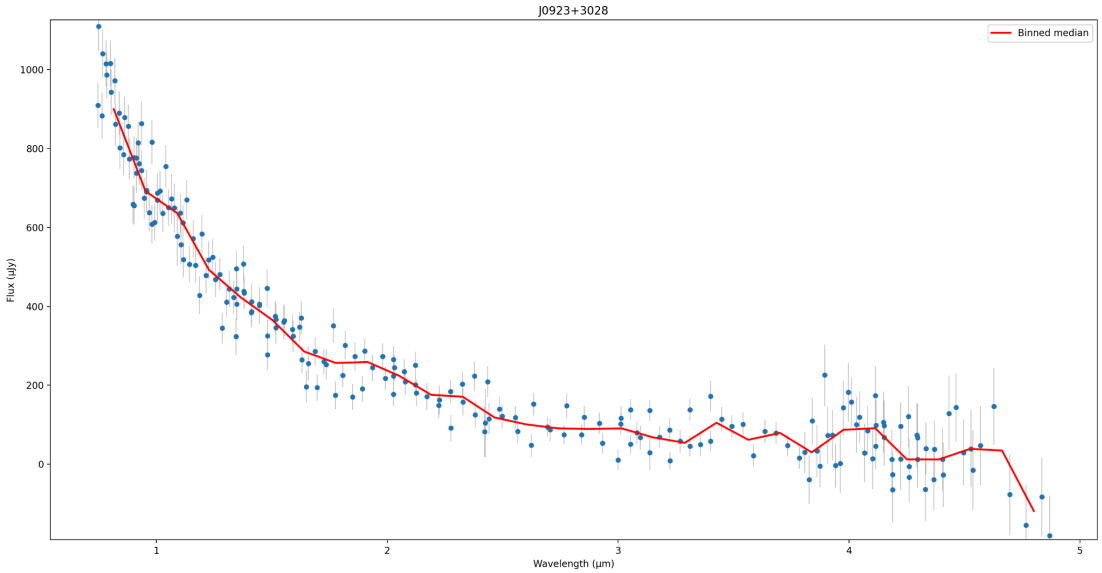
- Info from Aladin
  - Gaia DR3 70034692, PM:29.885
  - Gaia RA and DEC: 140.93999002815 30.46794693363
  - Main ID from Simbad catalog: CBS 6, PM: pmra-3.797 pmdec-29.642
  - Sinbad RA and Dec: 140.9400096070 +30.4680786779
- Simbad Coord website
  - ID: CBS 6
  - PM: **-3.797 -29.642**
- Using python and the plotting with IRSA
  - Calculated RA (in degrees): 140.939972
  - Calculated Dec (in degrees): 30.467815



- Compared to Gaia

12) J0923+3028

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



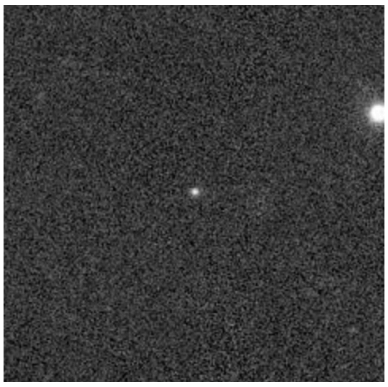
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down. Probably the most similar to the ISRA plot with the slope down.

Name Gmag Gmag_e ra dec

13) J1436+5010

18.4158 0.0014 14:36:33.303 50:10:26.843

- One big star on the edge of the image and it seems a small spot is a little brighter to the right of the star

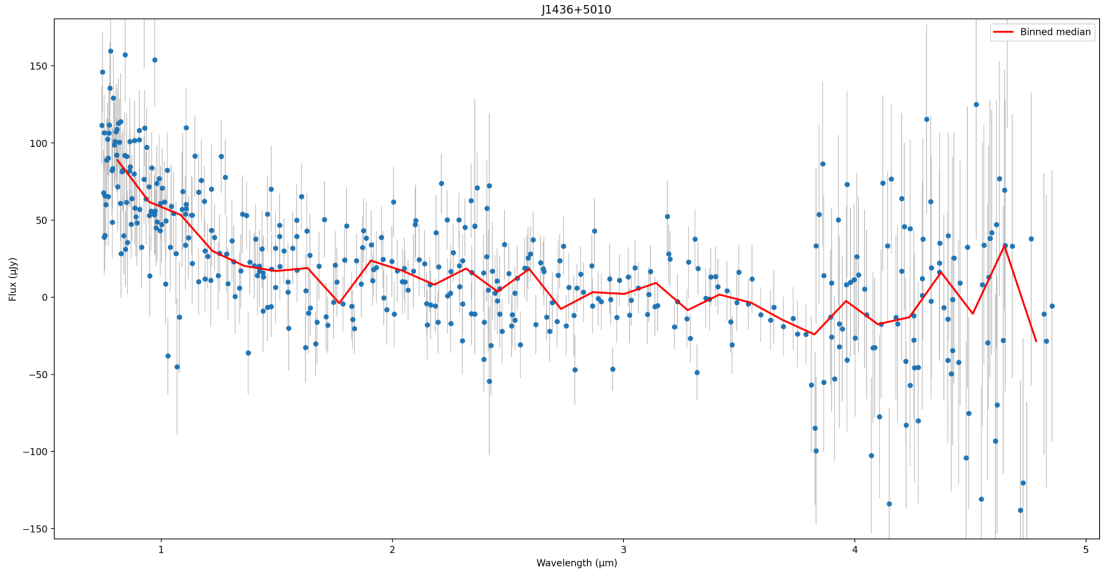


- Info from Aladin
  - Gaia DR3 16035547, PM:9.983
  - Gaia RA and DEC: 219.13876457169 50.17412295411
  - Main ID from Simbad catalog: SBSS 1434+503, PM: pmra8.558 pmdec-5.140
  - Sinbad RA and Dec: 219.1387051814 +50.1741457967
- Simbad Coord website
  - ID: SBSS 1434+503
  - PM: **8.558 -5.140**
- Using python code
  - Calculated RA (in degrees): 219.138822
  - Calculated Dec (in degrees): 50.174100
- Compared to Gaia

Photometry plot found on ISRA to be at 0?

13) J1436+5010

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



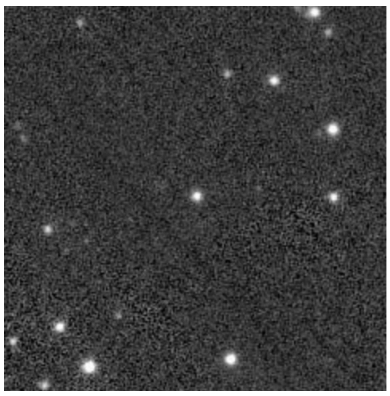
The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Name Gmag Gmag_e ra dec

14) J1832+2031

17.6301 0.0007 18:32:36.539 20:31:08.202

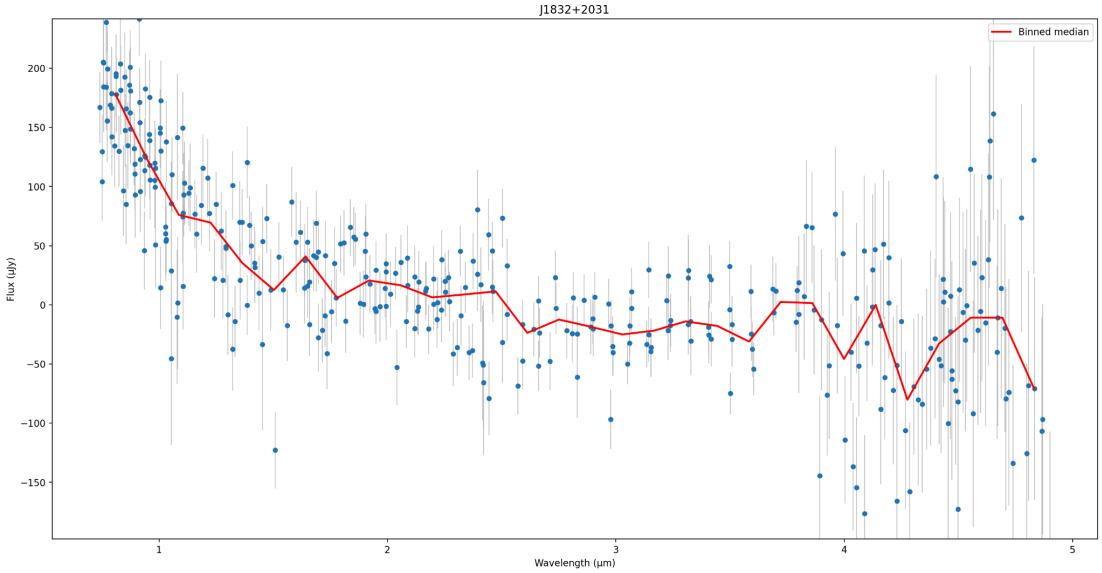
- A lot of other stars in the image



- Info from Aladin
  - Gaia DR3 45258721, PM:8.123
  - Gaia RA and DEC: 278.15224762656 20.51894493645
  - Main ID from Simbad catalog: SDSS J183236.53+2, PM: pmra8.116 pmdec-0.329
  - Sinbad RA and Dec: 278.1522091122 +20.5189463972
- Simbad Coord website
  - ID: SDSS J183236.53+2
  - PM: **8.116 -0.329**
- Using python code
  - Calculated RA (in degrees): 278.152284
  - Calculated Dec (in degrees): 20.518944
- Compared to Gaia

14) J1832+2031

Photometry plot made from ISRA data tables and constrained the flux error and flux so cut out the extreme outliers that made the photometry plot on ISRA hard to analyze.
X-axis: Flux (μJy) Y-axis: Wavelength (μm)



The red line shows the median of the flux over the wavelength of the light curve in the infrared. Shows the expected slope down.

Plot code notes:

Wavelength range (wl_range) = (0.7,5.0)

- Keeps only the instruments wl range

Flux error percent (err_pct) = 90 drops the noisiest 10% of the the table, so the threshold differs between sources

Flux error max(err_max) = 50 uses one fixed value for everything, which is better for comparing targets

The percentile is computed on *err[good]*, so bad rows don't skew it

Sigma clipping

- The wl range is split into *n_bins*, and outliers are flagged within each bin separately
- Dins with fewer than *min_per_bin* points are skipped, since statistics on a handful of points are unreliable.
- *Sigma clip*
    - Repeats up to 5 times: compute median and standard deviation, flags points beyond *clip_sigma* standard deviation, recompute, repeat.
- *.mask is True*
    - For rejected points, so ~ flips it into a keep-mask
- It only looks at **flux**, not error bars, and it compares points across att epochs in a bin
    - A source that really varies in time can lose real data

Can set *clip_sigma = None* to turn off clipping